

Mahi Singh

+91 6386699661 | mahisinghh9597@gmail.com | [linkedin.com/in/mahi-singh-ai](https://www.linkedin.com/in/mahi-singh-ai) | github.com/Mahi-S83

Education

VIT Bhopal University

Bhopal, India

Bachelor of Technology in Computer Science and Engineering — CGPA: 9.21 / 10.00

Aug 2023 – May 2027

Mercy Memorial School | ISC (Class XII): **89.75%** | ICSE (Class X): **95.40%**

Uttar Pradesh, India | 2023 & 2021

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C++, HTML/CSS

AI & Machine Learning: TensorFlow, PyTorch, YOLOv11, Faster R-CNN, Computer Vision, Deep Learning, Google Gemini, Vision API, Prompt Engineering, Pandas, NumPy

Web: Next.js, React, FastAPI, Node.js, Express.js, REST APIs, Tailwind CSS

Databases & Cloud: PostgreSQL, Supabase, MongoDB, MySQL, Vercel, Render, Railway

Developer Tools: Git, GitHub, Jupyter Notebook, VS Code, CI/CD, Agile, SDLC | **Spoken:** English, Hindi

Experience

AI Engineer Intern

Mar 2025 – Sep 2025

AmasQIS.ai

Oman (Remote)

- Engineered a real-time PPE compliance detection system using Python, TensorFlow, and Faster R-CNN to process live multi-camera factory feeds, achieving **90%+ precision** on the validation set.
- Developed an end-to-end object detection pipeline to classify safety violations (helmets, vests, gloves) across concurrent camera streams for active factory floor monitoring.
- Optimized model inference for **low-latency production deployment**; collaborated with cross-functional engineering teams on iterative model improvements through structured optimization cycles.

Projects

JeevanTrack — AI Health Timeline Platform | *Next.js, FastAPI, Supabase, Gemini 2.5, TypeScript*

Live App | GitHub

- Built an AI-powered health record platform** that digitizes, summarizes, and visualizes patient medical history into a unified timeline using Gemini Vision, Next.js, FastAPI, and Supabase.
- Architected an offline-first data layer using localStorage-first caching and cloud synchronization, eliminating data fragmentation across **5+ core platform features**.
- Integrated Google Gemini 2.5 Flash via Vision API to automate medical report text extraction, reducing manual data entry by **90%** and rendering concise 1-page clinical summaries in under **30 seconds**.
- Designed a resilient multi-tier fallback system (local storage → rules-based matching → manual flow) ensuring zero user-facing failures and **100% reliability** during upstream AI disruptions.

knowYourFood — GenAI Ingredient Analyzer | *React, FastAPI, Python, Gemini 2.0, PostgreSQL*

Live App | GitHub

- Developed a GenAI-powered food label analyzer** that uses Google Gemini to explain ingredient safety, identify potentially harmful additives, and generate consumer-friendly health insights in real time.
- Architected the platform with a React frontend and asynchronous FastAPI backend, achieving a **sub-3-second** average response time across **150+ product analyses**.
- Designed a secure REST API with structured prompt engineering, PostgreSQL authentication, and strict CORS security policies; implemented responsible AI output validation for safe, accurate responses.

KharpatNaashak — Automated Weed Detection System | *YOLOv11, PyTorch, OpenCV, Gradio, Python Model Space* | *GitHub*

- Developed a computer vision system for automated weed detection** in soybean fields using YOLOv11, enabling precise identification of crop weeds from field images.
- Trained a YOLOv11n model identifying **16 weed species**, achieving **60.7% mAP@0.5**, **67.1% precision**, and **52.2% recall** on the MH-Weed16 dataset (25,797 images).
- Built a bilingual Gradio web interface with video upload, density heatmaps, and a herbicide cost-savings calculator demonstrating a **60–70% chemical reduction** through localized herbicide targeting.

Leadership & Volunteer

Technical & Community Volunteer

Jul 2023 – May 2025

eVidyaloka · Pahal Foundation · Pledge A Smile

- Delivered virtual STEM education to **50+ underprivileged students** across rural India; improved assessment scores by **30%** and created digital fundraising campaigns for Pledge A Smile.

Certifications

– **Applied Machine Learning in Python** — Coursera, University of Michigan

Dec 2025

– **Introduction to Internet of Things** — NPTEL, IIT Kharagpur (*Elite + Silver* | *Score: 84%*)

Jan–Apr 2026

– **Cloud Computing** — NPTEL, IIT Kharagpur (*Elite* | *Score: 66%*)

Jan–Apr 2025